

Technical Datasheet Input / Output Modules with Modbus RTU Protocol with RS485 Interface

The IO modules communicate via RS485. The port can drive distances up to max 700 meters without the use of any repeater (*this feature however also depends on the signal strength of the Modbus Master Device*).

The RS485 Digital IO module is sturdy, has low power usage and is easy to use.

8 DI + 8 AI Module: -



The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators.

The design of the modules incorporates '**resettable Fuses**' to safeguard against reverse polarity connection both for **Power** and **Communication** port.

Specifications

General –

I/O Connectors	12 Pin 5.08 mm pitch pluggable screw Terminal Block, 2 Pin 5.08 mm pitch pluggable screw terminals.
Dimensions	110 mm L x 110 mm B x 50 mm H
Power	Input Power – 12 - 24 VDC or 24 V AC/DC Typical – 12V DC @ 120mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



DI Inputs –

Channels	8
Sense Voltage	3.3 – 30 VDC
Sense Logic	Logic '0' = <1 VDC, Logic '1' = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec

AI Inputs –

Channels	8
Input Signal	4 – 20 mA / 0 – 10V DC(jumper selectable)
Accuracy	± 2% Full scale
Input Resolution	10 bits / 12 bits (Optional)
Isolation	Optically Isolated
External Loop Voltage	+ 12 VDC min
AI input impedance	120Ω

Additional Features: -

All inputs and communication ports isolated
Input power reverse polarity safety
ESD Safety IEC 61000-4-2, ± 30KV contact, ± 30KV air
EFT IEC 61000-4-4, 50A (5/50ms)
750V isolation.
CRC Error check.
No configuration needed on the IO board

Configuration Settings: -

Communication Speed	9600 – 115200.
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave Id	Soft Setting
Function code DI	0x02 Read discrete Input
DI Register Address	0,1,2,3,4,5,6,7,8.
Function Code AI	0x03 Read Holding Registers
AI Register Address	10 Bit - 1,2,3,4,5,6,7,8
Soft Serial Function Code	0x03 Read Holding Registers
Soft Serial Register Address	11,12,13,14,15,16,17,18.

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
0	DI 1(With & Without Potential)	10001	0X02	RS485	1 Bit Boolean
1	DI 2(With & Without Potential)	10002	0X02	RS485	1 Bit Boolean
2	DI 3(With & Without Potential)	10003	0X02	RS485	1 Bit Boolean
3	DI 4(With & Without Potential)	10004	0X02	RS485	1 Bit Boolean
4	DI 5(With & Without Potential)	10005	0x02	RS485	1 Bit Boolean
5	DI 6(With & Without Potential)	10006	0x02	RS485	1 Bit Boolean
6	DI 7(With & Without Potential)	10007	0x02	RS485	1 Bit Boolean
7	DI 8(With & Without Potential)	10008	0x02	RS485	1 Bit Boolean
1	AI 1 – 10 Bit	40002	0X03	RS485	16 Bit Unsigned int
2	AI 2 – 10 Bit	40003	0X03	RS485	16 Bit Unsigned int
3	AI 3 – 10 Bit	40004	0X03	RS485	16 Bit Unsigned int
4	AI 4 – 10 Bit	40005	0X03	RS485	16 Bit Unsigned int
5	AI 5 – 10 Bit	40006	0x03	RS485	16 Bit Unsigned int
6	AI 6 – 10 Bit	40007	0x03	RS485	16 Bit Unsigned int
7	AI 7 – 10 Bit	40008	0x03	RS485	16 Bit Unsigned int
8	AI 8 – 10 Bit	40009	0x03	RS485	16 Bit Unsigned int
18	Display Baud Rate (Default:960)	40019	0x03	RS485	16 Bit Unsigned int
19	Enter New Baud Rate	40020	0x03	RS485	16 Bit Unsigned int
20	Display Slave ID (Default:1)	40021	0x03	RS485	16 Bit Unsigned int
21	Enter New Slave ID	40022	0x03	RS485	16 Bit Unsigned int

22	Display Parity (Default: None-3And for Odd-1,Even-2)	40023	0x03	RS485	16 Bit Unsigned int
23	Enter New Parity	40024	0x03	RS485	16 Bit Unsigned int.
24	Display Stop Bit (start-2/stop bit-1)	40025	0x03	RS485	16 Bit Unsigned int.
25	Enter New Start/ Stop Bit.	40026	0x03	RS485	16 Bit Unsigned int.

Note: -

1. In register 40023- you will be able to see the current parity and you can change it to register 40024
2. In register 40025- you will be able to see the current stop bit, and you can change it in register 40026
3. If you want to enter the baud rate 9600/19200/38400/57600/115200etc.
Enter it as 960/1920/5760/11520etc.
4. After that press the reset button or reboot the system.

BAUD RATE and SLAVE ID DESCRIPTION:

Tx = 103: Err = 4: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010	Alias	00020
0		0		0		1
1		0		0		0
2		0		0		3
3		0		0		0
4		0		0		1
5		-2731		0		0
6		-2731		0		0
7		-2731		0		0
8		-2731		960		
9		0		0		

As you can see in the image above, the resistor 40019 – 960 represents the present baud rate (9600) of the board and 40021– 1 represents the present slave id of the board.

To change the baud rate and slave id you must enter a value on the 40020 resistor for the baud rate and 40022 for the slave id as shown in the image below.

Read/Write Definition

Slave ID: 1

Function: 03 Read Holding Registers (4x)

Address: 20

Quantity: 11520

Scan Rate: 1000 ms

☒ Read/Write Enabled

Read/Write Once

View

Rows: ☒ 10 ☐ 20 ☐ 50 ☐ 100

☐ Hide Alias Columns

☐ Address in Cell

Display: Signed

☐ PLC Addresses (Base 1)

Read/Write Definition

Slave ID: 5

Function: 03 Read Holding Registers (4x)

Address: 0

Quantity: 10

Scan Rate: 1000 ms

☒ Read/Write Enabled

Read/Write Once

View

Rows: ☒ 10 ☐ 20 ☐ 50 ☐ 100

☐ Hide Alias Columns

☐ Address in Cell

Display: Signed

☐ PLC Addresses (Base 1)

Tx = 185: 4: ID = 1: F = 03: SR = 1000ms

Save

	Alias	00000	Alias	00010	Alias	00020
0		0		0		1
1		0		0		5
2		0		0		3
3		0		0		1
4		0		0		1
5		-2731		0		0
6		-2731		0		0
7		-2731		0		0
8		-2731		960		
9		0		11520		

As you can see in the image above I set 115200 baud rate and slave ID 5.

Note:

1. If you want to enter the baud rate 9600/19200/38400/57600/115200etc.
Enter it like this 960/1920/5760/11520etc.
2. After that press the reset button or reboot the system.
3. Then you will get the present value of 11520 at 40019 baud rate(115200) and 5 at 40021 slave id.
4. You can Change Parity Bit , stop bit Of the device with same way.

Tx = 9: Err = 3: ID = 5: F = 03: SR = 1000ms

	Alias	00000	Alias	00010	Alias	00020
0		0		0		5
1		0		0		0
2		0		0		1
3		0		0		0
4		0		0		1
5		-2731		0		0
6		-2731		0		0
7		-2731		0		0
8		-2731		11520		
9		0		0		

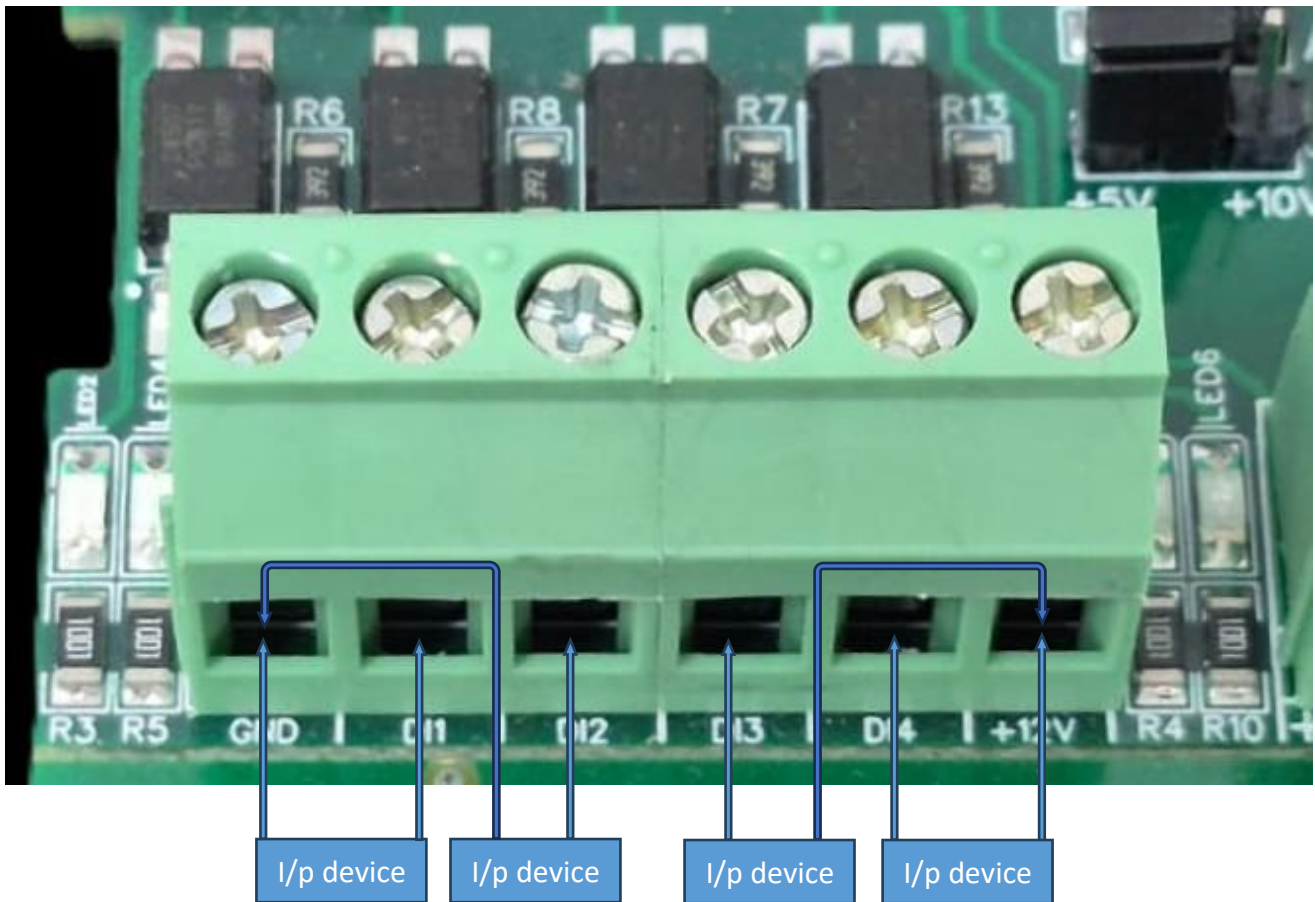
Note: -

For MODBUS communications, a **shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

Recommended Cable Electrical Characteristics: -

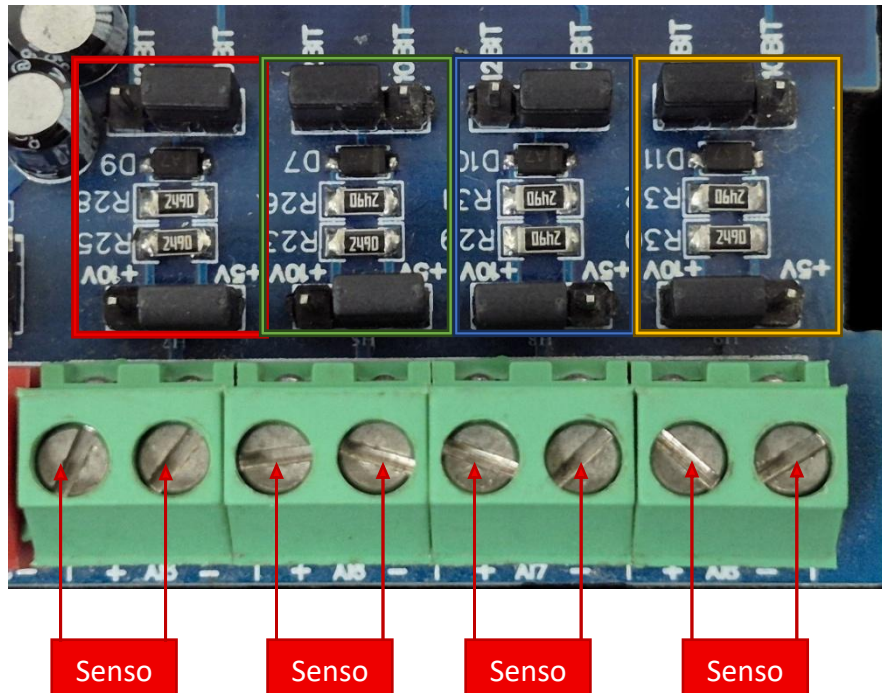
22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

DI



- **Operation:** The module provides four isolated input channels (**DI1–DI4**) for sensing external signals or dry contacts. These inputs are opto-isolated for circuit protection.
- **Terminal Layout:**
 - **GND:** Common Ground reference.
 - **DI1 – DI4:** Individual Digital Input channels.
 - **+12V:** Internal reference voltage for sourcing "Dry Contact" switches.
- **Wiring Options:**
 - **Dry Contact:** Connect a switch between the **+12V** terminal and any **DI** pin.
 - **External Voltage:** Connect the external signal to a **DI** pin and the external ground to the **GND** terminal.
- **Voltage Rating:** The input circuitry is rated for a maximum of **80VDC**. Ensure external signals do not exceed this limit to prevent damage to the isolation ICs.

AI



- **Analog Input (AI) Operation:** Each channel provides two terminals: + (Positive) and - (Negative/Common). These are used to read external analog sensor signals.
- **Channel Configuration (Visual Guide):** Each of the four AI channels features two dedicated jumpers to independently set the voltage range and bit resolution. Refer to the color-coded boxes in the board diagram to configure your desired scenario:
 - **Red Box (5V, 10-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **10 BIT**.
 - **Green Box (5V, 12-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **12 BIT**.
 - **Blue Box (10V, 10-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **10 BIT**.
 - **Yellow Box (10V, 12-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **12 BIT**.
- **Sensor Wiring Connections:**
 - The analog sensor's positive signal wire should be connected to the + terminal of the desired channel.
 - The sensor's common or ground wire should be connected to the corresponding - terminal.
- **Crucial Setup Recommendation:** Always ensure that the voltage jumper setting strictly matches the maximum output range of your connected sensor (either 5V or 10V) to guarantee accurate analog-to-digital conversion and prevent data clipping.
- **Crucial Setup Recommendation:** Always ensure that both the mode jumper (NTC vs. AI) and the voltage jumper strictly match the hardware you are connecting. Incorrectly setting an NTC to an AI mode, or mismatching the voltage limits, will result in inaccurate temperature or signal readings.

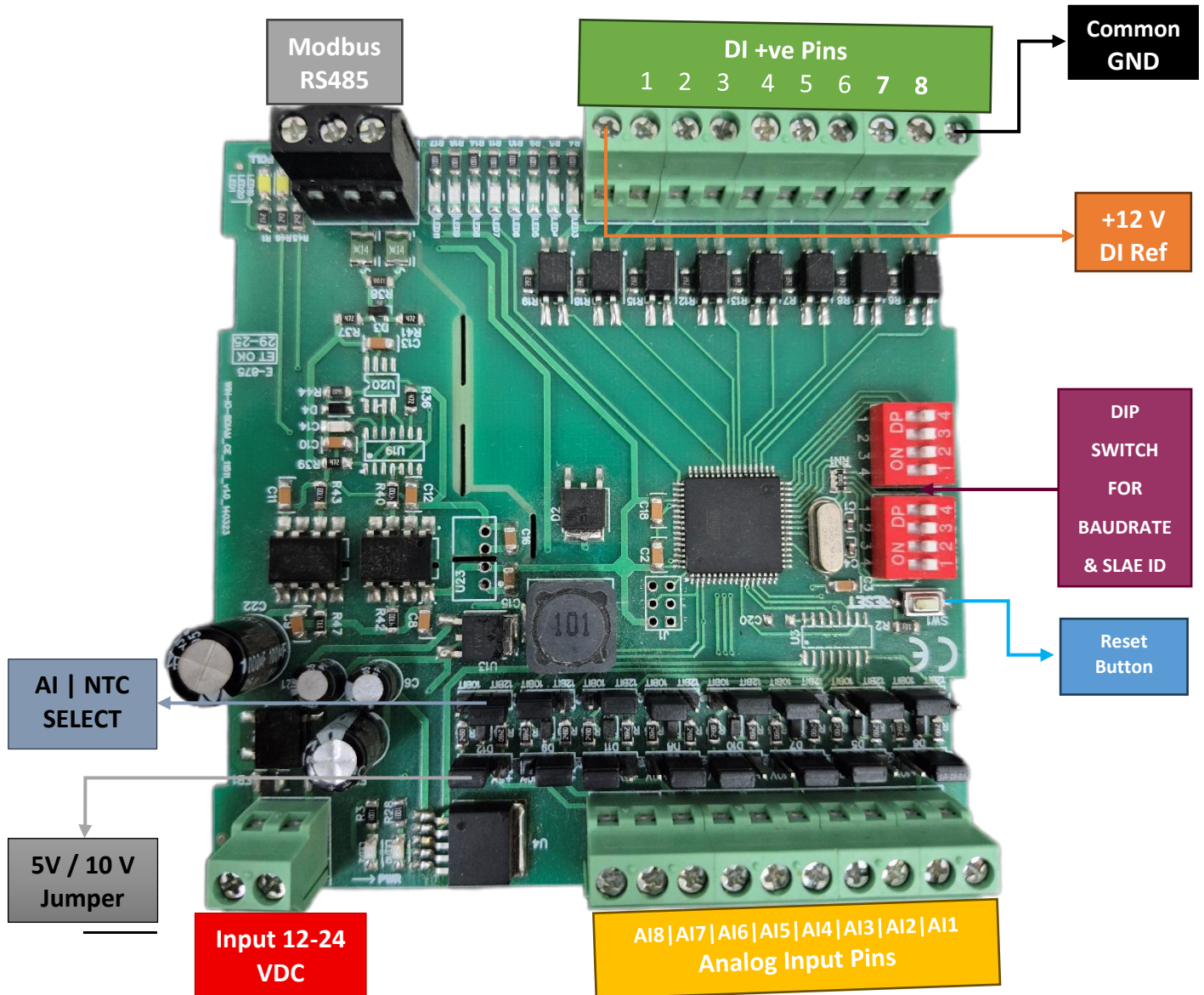
BAUD RATE DESCRIPTION



Default Setting
DIP switch

NOTE –

- By default, the Jumper setting on the board is 10 V and 10 Bit.
- By default, Baud rate DIP Switch 1 is set to ON, toggling it OFF and pressing reset button will bring the device to default RS485 Configuration i.e. Baud rate - 9600, Stop bit - 1, Parity - None, Data Bit - 8. Keep the indicated DIP switch to ON position for setting desired configuration.



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